

TASK 02 — PART B: DIAGNOSIS

Module 02 · Week 02 — Management of Information Systems (IS) Assessment

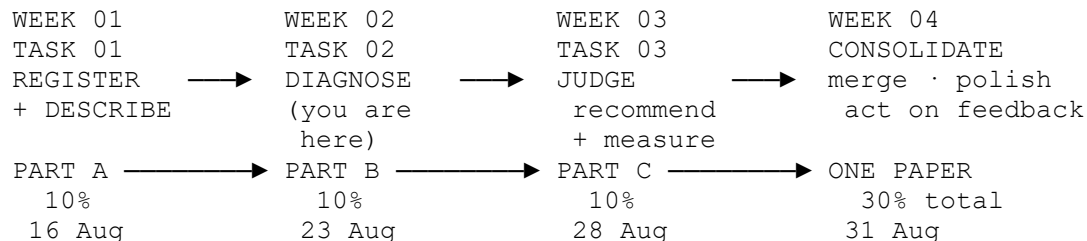
Cyber.SoHo Educational Hub — IT 500 (2025–2026) *From the Industry to the Classroom — Building the IT's Future, Today and Together!*

Weight	10% of IT 500 · instalment 2 of 3 in the Written Project (30% total)
Produces	Part B — Diagnosis, 1,000–1,200 counted words
Working time	3 hours maximum. If you pass 3h30, stop and submit what you have
Due	Midnight CET, Sunday 23 August 2026
Where	Blackboard → Assignments → DropBox. <i>Emailed submissions are not submissions</i>
Subject	The same topic, organisation and system you registered in Task 01. No switching

0. BEFORE YOU START — READ ONCE (4 minutes)

0.1 Where this sits: the capstone, not a homework

You are not writing three essays. You are writing **one 3,000-word academic paper in three instalments**, and this is the middle one. Task 01 described your system. Task 02 takes it apart. Task 03 will judge it.



The single most valuable thing you own right now is **your Task 01 feedback**. Open it before you write a word of Part B. Students who act on instalment-one feedback typically gain several marks on the final paper; students who do not, hand in the same three mistakes three times.

Two things carry forward from Task 01 and are marked again here: your evidence log (every claim, traceable to where you got it) and your A7 open questions (the things you could not answer). Part B exists largely to close some of those questions.

0.2 Your three hours, spent on the course ponderation (2-1-3)

This course runs on a **2-1-3 ponderation** — 2 hours of concept, 1 hour of hands-on lab, 3 hours of independent production, each week. Task 02 uses the same 2 : 1 : 3 proportion to divide your three-hour budget. Set a timer for each block. When a block's time is gone, move on with whatever you have — an incomplete pass that is written down scores; a perfect pass that never got written scores nothing.

Block	Ponderation share	Minutes	What you do	What comes out
1 — CONCEPT	2	60	Four diagnostic passes (B1–B4) over your own system	Four short findings, ~40 words each
2 — LAB	1	30	Put <i>your</i> numbers into the Week 02 lab files	Four numbers you can defend
3 — WRITE	3	90	Findings + six-lens rescore + open questions	Part B, 1,000–1,200 words, .docx

0.3 Vocabulary, for the first time you meet it

You are computer science students, but you are new to **IS (Information System)** assessment. Every term below appears in this task. Fuller definitions are in the Week 02 glossary spreadsheet.

Term	Plain meaning
IT (Information Technology)	The equipment: hardware, software, networks
IS (Information System)	The equipment plus the people and the process around it. Assessment is always of the IS, never of the IT alone
System of record	The one place where a fact is <i>supposed</i> to live authoritatively
Handoff	A point where work passes from one person, team or system to another. Handoffs are where time and information are lost
Data duplication	The same fact stored in more than one place, free to disagree with itself
Integration	Two systems exchanging data automatically. If a human retypes it, that is not integration — that is a handoff
API (Application Programming Interface)	An agreed way for one program to ask another program for something

Term	Plain meaning
ETL (Extract, Transform, Load)	Pulling data out of one system, making it consistent, loading it into another
DBMS (Database Management System)	The software that stores data and refuses data that breaks its rules
Primary key / Foreign key	The column that identifies a row uniquely / the column that points at another table's primary key
SQL (Structured Query Language)	The language used to ask a database questions
TCO (Total Cost of Ownership)	Every cost over the life of a decision, not the purchase price
IaaS / PaaS / SaaS	Infrastructure / Platform / Software as a Service . Three rental models, distinguished by who is responsible for which layer
Bandwidth vs latency	How much data moves per second (Mbps) vs how long one packet takes to arrive (ms). Independent quantities. Buying more of one never fixes the other
Airtime contention	On Wi-Fi (Wireless Fidelity), only one device per channel transmits at a time. The rest wait. Invisible on a speed test
CIA triad	Confidentiality, Integrity, Availability — the three properties security protects
SLE × ARO = ALE	Single Loss Expectancy × Annualized Rate of Occurrence = Annualized Loss Expectancy: what a risk costs per year
ROSI (Return on Security Investment)	Whether a control returns more than it costs. Negative means do not buy it
Defence in depth	Seven layers of protection, on the assumption that every one of them will eventually fail

0.4 Rules of engagement — these do not bend

They applied in Task 01 and they apply for the whole term.

1. Gather evidence **as an ordinary user**, or from **published sources**. Nothing else.
2. Collect **no personal data** about anyone. No names, no records, no screenshots containing another person's information.
3. **Test nothing**. No scanning, no probing, no credential attempts, no "just checking whether it's vulnerable". Observation only.
4. **Never contact** a clinical, emergency, non-emergency or dispatch line for coursework. Healthcare and Public Safety topics are worked from published sources exclusively.
5. **Publish nothing**. Your findings stay in your submission.
6. If a step in this task cannot be done lawfully or ethically for your chosen system, **write that down as a finding and move on**. "I could not observe this without breaching rule 2" is a legitimate, markable answer. Inventing the observation is not.

1. THE BRIEF, IN ONE SENTENCE

Take the system you described in Task 01, look underneath it at four specific layers — infrastructure, data, network, security — and report **where information is duplicated, where it is lost, and what the handoffs actually cost**, with a measurement beside every claim.

Part A asked *what is this?* Part B asks *what is wrong with it, and how do you know?*

BLOCK 1 — CONCEPT (60 minutes) · four diagnostic passes

Fifteen minutes each. Set a timer. Each pass produces **one written finding of roughly 40 words** plus the evidence for it. A pass with a score and no written finding **scores zero** — that rule has not changed since Task 01.

For every pass, if you genuinely cannot obtain the evidence lawfully, write the *reasoned estimate* instead and label it clearly: [ESTIMATE — basis: ...]. Marked honestly, an estimate loses very little. An unlabelled guess loses everything.

B1 · The infrastructure pass (15 min) — *IT Infrastructure and Emerging Technologies*

Question: what does your system run on, who is responsible for which layer, and what does that arrangement cost over five years?

Steps

1. Decide which rental model your system uses: **on-premises, IaaS, PaaS or SaaS**. If you cannot tell from published sources, say which two it lies between and why.
2. Draw the responsibility split. For each of these layers — application, runtime, operating system, virtualization, servers and storage, networking, **and your organisation's data** — write "**organisation**" or "**provider**".
3. Answer the question that decides the pass: **which layer, if it failed, would the organisation still be accountable for even though the provider operates it?** In almost every case the answer is the data row, and stating *why* is the mark.
4. Build a five-year TCO (Total Cost of Ownership) sketch using the Block 2 lab file. Two options only: the arrangement they have, versus one plausible alternative.
5. **Include staff time in the model.** Hours per week × weeks per year × an hourly rate you can justify. This is the input most often omitted and it is usually the largest single variable.

Finding to write (~40 words): state the model, name the layer where responsibility and accountability diverge, and give the five-year figure with the labour rate you assumed.

B2 · The data pass (15 min) — *Foundations of Business Intelligence: Databases and Information Management*

Question: where does the data live, where is the same fact stored twice, and what question can this system *not* answer?

Steps

1. List the **entities** your system handles — the things it stores facts about. Three to six is normal (for a ticketing system: requester, ticket, asset, technician).
2. For each entity name the **system of record**: the one place the fact is supposed to live authoritatively.
3. Hunt for duplication. Ask: *is any of this typed in twice?* Look for a fact that exists in the main system **and** in a spreadsheet, an email, a form, or a second system. Name at least one.
4. State the consequence in the vocabulary of Week 02: which of the five diseases of a file environment is present — redundancy, inconsistency, program–data dependence, poor security, or lack of data sharing?
5. Name **one question a manager would reasonably ask that this system cannot answer**, and say precisely which missing key, missing field or missing link makes it unanswerable. This is the highest-value sentence in B2.

Finding to write (~40 words): one duplicated fact, its consequence named in Week 02 vocabulary, and one unanswerable management question with the structural reason.

B3 · The network pass (15 min) — *Telecommunications, the Internet, and Wireless Technology*

Question: how does a request travel through this system, and where does the delay or the loss actually sit?

Steps

1. Trace **one** transaction end to end — a booking, a ticket, a scan, an order. Write the hops in order: user device → access network → internet → provider → back. Six to nine steps.
2. Mark every step where the work passes between people or systems. Those are your **handoffs**.
3. For your slowest or most fragile handoff, estimate two things separately: **bandwidth** (how much data, in Mbps) and **latency** (how long a round trip takes, in ms). Then say which one the complaint is actually about. Most complaints described as "slow" are latency or a handoff, not bandwidth.
4. If the system has a wireless component, ask whether **airtime contention** could explain the symptom before assuming a capacity shortage.

5. Cost the handoff in something other than money: **minutes**, **re-keying steps**, or **information lost in transit** (fields that exist on one side and not the other).

Finding to write (~40 words): name the constraint, distinguish it from the constraint people assume, and give the handoff cost in your chosen unit.

B4 · The security pass (15 min) — *Securing Information Systems*

Question: what would go wrong, how much would it cost per year, and is the obvious fix worth buying?

Steps

1. Name **one realistic incident** for your system. Realistic means: a phishing or social-engineering route, credential reuse, an unpatched component, or an insider error. Not a film plot.
2. Classify the damage against the **CIA triad**: for each of confidentiality, integrity and availability, write *breached* or *not affected*, with one clause of justification.
3. Identify which of the seven **defence in depth** layers would have to fail for your incident to succeed — physical, perimeter, network, endpoint, application, data, or people-and-process. Layer 7 is the answer far more often than students expect.
4. Quantify: estimate **SLE** (cost of one occurrence) and **ARO** (how many times per year), and multiply for **ALE**.
5. Propose **two** controls: one that costs money and one that costs only a process change. Compute **ROSI** for both: $(ALE_{before} - ALE_{after} - Cost) / Cost$. Report both, including if one is negative.

Finding to write (~40 words): the incident, the failed layer, the ALE, and which of your two controls the arithmetic actually supports.

Ethical boundary for B4, stated plainly. You are describing a *plausible* incident from published sources and ordinary observation. You are not probing, scanning, testing, or reporting a real vulnerability in a real organisation. If you believe you have accidentally discovered a genuine weakness, **do not write it in this assignment** — close your notes and contact your instructor privately.

BLOCK 2 — LAB (30 minutes) · your numbers, our files

You do not build tools this week. You **reuse the Week 02 lab files** and replace the fictional figures with your own estimates. Roughly seven minutes each.

Pass	Open this file	Change these inputs	Record this output
B1	data/W02_L1_TCO_Model.xlsx (or) code/L1_tco_calculator.py	Analysis period, hourly rate, working weeks, both options' up-front and recurring costs	Five-year total for each option, the difference, and the break-even year
B2	code/L2_club_inventory_db.py (or) data/W02_L2_Normalization.xlsx	Replace the three tables with your entities; keep the keys and the CHECK constraint	Which of your duplications the constraints would have prevented
B3	code/L3_network_workbench.html	Users, concurrency %, demand per user, link speed, access points, channels in use , devices per person	Link utilisation %, wireless utilisation %, and which one the tool names as the constraint
B4	code/L4_risk_workbench.html (or) data/W02_L4_Risk_Register.xlsx	Your SLE, your ARO, your two control costs	ALE, both ROSI figures, and the ranking

Two habits that earn marks here. First, when you change an input, **write down why that value and not another** — one clause is enough, and it is what turns a guess into an estimate. Second, run each tool **twice**: once with your realistic figure and once with a deliberately pessimistic one. If the recommendation survives both, say so; if it flips, that is a finding worth more than the original number.

The Python files run with no installation beyond Python itself. The two HTML tools open in any browser with the internet switched off. Nothing here needs an account.

BLOCK 3 — WRITE (90 minutes) · producing Part B

B5 · Duplication and handoff findings (~180 words)

Pull B2 and B3 together into the section the brief actually asks for. State, in connected prose:

- one place information is **duplicated**, and what disagreement that permits;
- one place information is **lost** — a field that exists on one side of a handoff and not the other;
- what the handoffs **cost**, in minutes, re-keying steps, or lost fields.

B6 · The six-lens rescore (~200 words) — where most of the marks live

Rescore the same six lenses you used in Task 01, **with your Part B evidence**. Ethics and Evidence count double. Every score needs a written finding beside it or that lens scores zero.

Lens	Question it answers	Weight	Task 01	Task 02	Movement — and why
1. Purpose	Which business objectives does the system serve?	×1			
2. People	Who inputs, who decides, who is affected?	×1			
3. Process	Which steps does it change, remove or create?	×1			
4. Strategy	Which competitive force does it weaken?	×1			
5. Ethics	Whose data, whose risk, whose accountability?	×2			
6. Evidence	What measurement proves each claim is true?	×2			

Report your weakest lens, not your average. Averaging conceals exactly the thing the assessment asks you to find. A paragraph that says *"Evidence remains my weakest lens at 2 of 5, because three of my B1 figures are labelled estimates"* scores considerably better than a comfortable average with nothing named.

Any lens whose score **moved** between Task 01 and Task 02 must be justified in one sentence. A score that moves without a reason reads as adjustment to taste.

B7 · Open questions (~60 words)

Two lists, both marked:

1. **Closed from A7.** Which Task 01 open questions did Part B answer? Quote the question, give the answer, cite the evidence.
2. **Newly opened.** What did this diagnosis raise that you cannot yet answer? These become the raw material for Task 03, so write them precisely enough to be actionable.

Word budget — keep yourself honest

Section	Words
B1 Infrastructure finding	150
B2 Data finding	180
B3 Network finding	150
B4 Security finding	180
B5 Duplication and handoff	180
B6 Six-lens rescore	200

Section	Words
B7 Open questions	60
Total	≈ 1,100 (target band 1,000–1,200)

Format — set it once, never touch it again

Microsoft Word .docx · **Times New Roman 12** · **double spaced** · **one-inch margins** on all four sides · cover page (title, your name, professor's name, course title, date) · **APA** citations and references · at least **three new academically sound sources** this week, carried into the running reference list · all figures, tables and screenshots **imported into the Word document**, never attached separately.

Cover page, references and appendices are **not counted** in the word total. The evidence log and the six-lens table go in the appendix.

Run **SafeAssign Direct** before you submit — not only on the final paper.

2. MARKING — 100 POINTS, SCALED TO 10%

#	Criterion	Points
1	B1 Infrastructure pass — model identified, responsibility split correct, five-year figure with labour included	12
2	B2 Data pass — entities and system of record, one duplication named, one unanswerable question with its structural cause	12
3	B3 Network pass — transaction traced, handoffs marked, bandwidth distinguished from latency, constraint named	12
4	B4 Security pass — incident, CIA classification, failed layer, ALE, two controls with ROSI	12
5	B5 Duplication and handoff cost — stated in a non-monetary unit, with evidence	10
6	B6 Six-lens rescore — all six present, findings beside every score, movements justified, weakest lens reported	20
7	B7 Open questions — A7 questions closed with evidence; new questions actionable	8
8	Evidence log — cumulative, every claim traceable, estimates labelled as estimates	8
9	Format and sources — Course Plan format met, APA correct, ≥3 new sound sources, SafeAssign run	6
	Total	100

Automatic zeros within a criterion: a score with no written finding; an unlabelled estimate presented as an observation; a claim in the evidence log with no traceable source.

3. A WORKED FRAGMENT (entirely fictional)

Subject: Calderbank University IT service-desk ticketing and escalation system. Every figure below is invented for teaching.

B2 finding. The requester's department is stored in the ticket record and again in the staff directory, and the two disagree for roughly one ticket in nine after any internal move (observed across 45 publicly visible ticket summaries over three weeks). This is data redundancy producing inconsistency. The consequence: the desk cannot answer "which department generates the most password resets?", because the department field on the ticket is a free-text copy rather than a foreign key to the directory. [ESTIMATE — basis: sample of 45 summaries; ratio extrapolated]

Notice four things: a fact, a number, the Week 02 vocabulary used correctly, and the estimate labelled. That is roughly 70 words and would score well. It contains no personal data and required no testing.

4. THE SUBSTITUTION CLAUSE — READ THIS, IT IS NOT DECORATION

Every tool, technique, file format and method named in this task is an example, not a requirement. All of it is open and fully substitutable.

If LibreOffice Calc, Google Sheets, Numbers, pandas, R, Julia, DuckDB, PostgreSQL, a Jupyter notebook, a whiteboard photograph or a hand-drawn table serves you better — **use it**. If you would rather write your own script than open our spreadsheet, that is a better answer, not a worse one.

One condition. If you substitute anything, add a short block to your appendix:

```
SUBSTITUTION NOTE
Original suggested : W02_L3_Bandwidth_Budget.xlsx
What I used instead: a Python script using pandas
Why                : I wanted to sweep 200 concurrency values rather than one,
                    and I already know pandas better than Excel formulas.
What changed in the result: none — same verdict at the default inputs.
```

Two clauses. It takes a minute, and it converts an unexplained deviation into evidence of judgement — which is precisely what the Evidence lens rewards.

Go further than this task asks. Research your own tools, compare **at least two** before choosing, and where nothing fits, **build one**. A student who writes forty lines of their own code

to answer a question our spreadsheet could not, and explains why, has demonstrated exactly the professional behaviour this course exists to produce. Say what you built and why, and it will be marked as a strength.

5. LINKS — every organisation and document named here

Resource	Why you would open it
University of Roehampton	The credential this project counts 100% toward
APA Style — official	Reference format required by the Course Plan
Purdue OWL — APA guide	Free worked APA examples when the official site is terse
NIST (National Institute of Standards and Technology)	Incident-handling phases and security control vocabulary for B4
ISO (International Organization for Standardization)	Standards referenced when describing management systems
IETF (Internet Engineering Task Force)	The protocol specifications behind your B3 trace
IEEE (Institute of Electrical and Electronics Engineers)	The 802.11 wireless standards family
W3C (World Wide Web Consortium)	Web standards, including accessibility, if your system is browser-based
OWASP (Open Worldwide Application Security Project)	Application-layer risk vocabulary for B4
CISA (Cybersecurity and Infrastructure Security Agency)	Published advisories usable as B4 evidence
Canadian Centre for Cyber Security	Canadian guidance and published threat assessments
ENISA (European Union Agency for Cybersecurity)	EU threat landscape reports, a strong APA source
Python · SQLite	The two tools the lab files use
MDN Web Docs	Reference if you extend the HTML workbenches
VirusTotal — URL scanner	Scan every external link through this before opening it

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6.7 Responsible disclosure. If, while observing as an ordinary user, you believe you have encountered a genuine security weakness in a real organisation, **do not document it in this**

assignment, do not investigate further, and do not tell your classmates. Close your notes and contact your instructor privately. Handling that situation correctly is itself professional conduct and will be recognised as such.

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